ASSIGNMENT 1

PAYMENT CALCULATOR FOR A VILLAGE MILK CO-OPERATIVE

TASK 12 – RE-EVALUATE AND ADAPT THE DESIGN

1. AIM

To re-evaluate the milk payment calculator and adapt its design to meet new requirements while maintaining accuracy, readability, and reusability.

2. OBJECTIVE

To analyse the impact of changing requirements, evaluate the original design, adapt the program, and justify the final design using execution results.

3. PART A – IMPACT ANALYSIS (2 MARKS)

The original milk payment calculator calculates the payment for one farmer at a time. The new requirement is to process multiple farmers in a single execution and calculate their combined payment.

This change affects the input handling, processing, and output sections of the program. The existing pricing formula remains unchanged. The calculate_payment() function can be reused to avoid rewriting the calculation logic.

4. PART B – EVALUATION OF THE ORIGINAL DESIGN (2 MARKS)

The original design uses constants, meaningful variable names, the Decimal data type, and a separate calculation function.

The use of Decimal improves accuracy in financial calculations. Constants make pricing rules easier to update, and functions improve readability and reusability.

However, the original design processes only one farmer at a time. Therefore, it needs to be extended to support multiple farmer records.

5. PART C – ADAPTATION OR REDESIGN (2 MARKS)

The design is adapted by storing multiple farmer records in a list and processing them using a loop. The existing calculate_payment() function is reused for each farmer.

The program calculates the individual payment for every farmer and adds the amounts to obtain the combined payment. This approach avoids code duplication and preserves the original pricing rules.

6. PART D – EVIDENCE-BASED DEFENCE (2 MARKS)

The adapted program was tested using three farmer records.

Farmer F001 supplied 10.5 litres of milk with 4.0% fat and 8.7% SNF. The calculated payment was ₹342.30.

Farmer F002 supplied 10 litres of milk with 3.5% fat and 8.5% SNF. The calculated payment was ₹300.00.

Farmer F003 supplied 5 litres of milk with 4.0% fat and 9.0% SNF. The calculated payment was ₹167.50.

The combined payment was ₹809.80 for three farmers.

These results demonstrate that the adapted design correctly calculates individual payments and the combined payment while reusing the original calculation function.

7. TIME AND SPACE COMPLEXITY

Time Complexity: O(n), because the program processes each farmer once.

Space Complexity: O(n), because the program stores the farmer records in a list.

8. RESULT

The milk payment calculator was successfully re-evaluated and adapted to process multiple farmers in one execution. The program calculates individual payments and the combined payment while maintaining the original pricing formula. The adapted design improves reusability, readability, and maintainability.

